Physiotherapy adherence

A physical bead counter instead of an app with streaks. Project name: Rep.

Sector: MedTech, physiotherapy adherence

Status: concept study, written and designed by Karthik Bindumon, 2026. Numbers are targets or modelled assumptions and are labelled as such.

Method: The Realisation Loop, v1 (see 00-realisation-loop.pdf).

1 Brief

Physiotherapists prescribe home exercise programmes and studies commonly report that fewer than half of patients complete them. Adherence apps respond with streaks, badges and reminders. For a person in pain, a broken streak reads as shame, and shame ends adherence. This study designs a physical, unconnected adherence object built as a ritual rather than a game.

2 Audience and needs analysis

Primary audience: outpatients with a six-week home programme after a musculoskeletal injury, seen by a physiotherapist every one to two weeks. What they know: the exercises, after one demonstration. What they lack: a reason to start a session when it hurts and a way to show the physio what actually happened. Constraints: pain, fear of re-injury, low trust in apps, shared households.

Needs analysis method: interviews with two physiotherapists and six patients, review of the clinic's current adherence sheet, and a diary study with three patients over one week.

3 The realisation

> **A person in pain needs a ritual, not a leaderboard. Counting beads is a ritual whole cultures already trust.**

Contrarian core: remove the app. The badge-and-streak model was built for people who are bored; it fails first for people who are hurt.

4 Learning objective (target)

Over six weeks, patients using Rep report completing at least 70 percent of prescribed sessions, against usual care, with no worsening in pain scores.

5 Design

Rep. A loop of ten wooden beads on a cord with a knot and one sliding marker. The physio sets the prescription in clinic: how many sets, how many reps per bead. One rep, one bead. When the marker comes home to the knot, the set is done. The loop lives where the exercise happens (bed post, chair, wrist).

Session scroll. A strip of paper on the fridge or wall; one tick per session. The patient brings it to each visit. The physio reads the scroll and the beads together and adjusts the prescription.

Family loop. Optional second loop for a household member who does the sets alongside. Relatedness without an app.

Implicit learning. The patient learns pacing from the object: a set has a beginning, a middle and an end you can feel. A missed day changes nothing about the object.

Artefacts. Bead loop spec (bead size, cord, marker), prescription card, scroll template, a one-page physio guide.

6 Theory map

Source	How it is used
Deci and Ryan, SDT	Autonomy: the patient sets the pace. Competence: the marker comes home. Relatedness: the physio reads the beads; the family loop.
Vlaeyen and Linton, fear-avoidance model	Graded exposure with no punishment for a missed day, so avoidance never gains a reason.
Winnicott, transitional objects	An object that holds the relationship with the physio between visits.
Mala practice	Counting as a calming act in Indian traditions (108 beads). Rep borrows the form at ten.
Fogg, anchoring	The loop sits where the exercise happens; the place is the prompt.
Schultz, reward prediction error	Small, predictable completion rather than a surprise badge.

7 Evaluation plan (Kirkpatrick)

Level	Measure	When
1 Reaction	One question: did the loop get in the way?	Each visit
2 Learning	Patient demonstrates a full set unprompted, with the loop.	Visit 2
3 Behaviour	Scroll ticks and bead state, corroborated at visits, against the clinic's usual adherence sheet.	Weekly
4 Results	Patient-Specific Functional Scale and pain score at six weeks, against usual care.	Week 6

8 Pilot plan and risks

One clinic, two physiotherapists, 20 patients, matched against the clinic's last 20 on usual care. Risks: the object read as childish (finish quality matters; wood, not plastic); lost loops (a second is issued without comment); self-report inflation on the scroll (the physio corroborates in the room).

9 Ethics

No data leaves the room by default; the object is not connected. Pain is respected: nothing about the object records a missed day. The family loop is the patient's choice.

10 What I would need from the client

Two physiotherapists for six weeks, access to the existing adherence sheet, ethics sign-off for a service evaluation, and a small budget for twenty-five loops.